

Experiment 5.2

Determination and plotting of Potential Difference for Positive Charge (Fixed r_1)

Code File Name: Exp5_2_PotentialDifference_Fixed_r1.m

Code

```
clc;
clear;
close all;

epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);

q = 1e-6;          % Charge (C)
r1 = 1;            % Fixed distance (m)
r2 = linspace(1.1,10,100); % Varying distance (m)

V = k*q*(1/r1 - 1./r2);

figure;
plot(r2,V,'LineWidth',2);
grid on;

xlabel('r_2 (m)');
ylabel('Potential Difference (V)');
title('Potential Difference for Positive Charge (Fixed r_1)');
```

Output Graph

